University Payroll Management System

A beginner-friendly command-line **University Payroll Management System** developed in Python for the Python Essentials course.

Project Overview

This project helps a university manage basic employee payroll operations from a terminal. It supports both **Faculty** and **Staff** employees and calculates salary, overtime, tax, retirement deductions, and net salary.

The application does not require a graphical interface or third-party Python packages.

Features

The program provides the following functions:

1. **Add Employee**
• Enter employee name
• Enter department
• Enter job role
• Select Faculty or Staff
• Enter monthly salary

2. **View Employees**
• Display all employees currently stored in the program.

3. **Search Employee**
• Search employees using their name or department.

4. **Add Overtime**
• Add overtime hours to an existing employee.

5. **Generate Payslip**
• Display a detailed salary calculation for an employee.

6. **Payroll Report**
• Display payroll information for all employees.
• Calculate total gross salary, tax, retirement deduction, and net salary.

7. **Remove Employee**
• Remove an employee using the employee ID.

8. **Exit**
• Close the application.

Payroll Calculation

The project uses the following simple rules:

Hourly Rate

```
Hourly Rate = Monthly Salary / 160
```

Overtime Pay

Overtime is paid at 1.5 times the normal hourly rate.

```
Overtime Pay = Overtime Hours × Hourly Rate × 1.5
```

Gross Salary

```
Gross Salary = Basic Salary + Overtime Pay
```

Tax

Faculty members:

```
Tax = Gross Salary × 10%
```

Staff members:

```
Tax = Gross Salary × 8%
```

Retirement

```
Retirement = Gross Salary × 5%
```

Net Salary

```
Net Salary = Gross Salary - Tax - Retirement
```

Technologies Used

• Python 3

• Python functions

• Lists

• Dictionaries

• Loops

• Conditional statements

• Exception handling

• User input/output

• Basic arithmetic

No external packages are required.

Requirements

• Python 3.8 or later

• Terminal / Command Prompt

The project is designed to run completely from the command line.

Installation

Step 1: Install Python

Install Python 3 from the official Python website if it is not already installed.

Verify the installation:

```
python --version
```

On some systems, use:

```
python3 --version
```

Step 2: Download or clone the project

Place the Python file in a folder.

For example:

```
university_payroll/
■■■ university_payroll.py
```

Step 3: Open a terminal

Navigate to the project folder:

```
cd university_payroll
```

How to Run

Run:

```
python university_payroll.py
```

Or, on systems where Python 3 is invoked using `python3`:

```
python3 university_payroll.py
```

The main menu will appear:

```
==========================================
 UNIVERSITY PAYROLL MANAGEMENT SYSTEM
==========================================
1. Add Employee
2. View Employees
3. Search Employee
4. Add Overtime
5. Generate Payslip
6. Payroll Report
7. Remove Employee
8. Exit

Enter your choice:
```

Example Usage

Add an employee

Choose:

```
1
```

Then enter information such as:

```
Enter employee name: Ananya Sharma
Enter department: Computer Science
Enter job role: Professor

Employee Type:
1. Faculty
2. Staff

Enter choice: 1
Enter monthly salary: 85000
```

The system creates an employee ID.

Add overtime

Choose:

```
4
```

Enter the employee ID and overtime hours.

Generate payslip

Choose:

```
5
```

Enter an employee ID.

The program displays:

```
================================================
              UNIVERSITY PAYSLIP
================================================
Employee ID      : 1
Employee Name    : Ananya Sharma
Department       : Computer Science
Role             : Professor
Employee Type    : Faculty
------------------------------------------------
Basic Salary     : ■ 85000.00
Overtime Pay     : ■ 0.00
Gross Salary     : ■ 85000.00
Tax              : ■ 8500.00
Retirement       : ■ 4250.00
------------------------------------------------
NET SALARY       : ■ 72250.00
================================================
```

Input Validation

The program checks several invalid inputs:

- Empty employee name
- Empty department
- Empty job role
- Invalid employee type
- Invalid salary
- Negative salary
- Invalid employee ID
- Invalid overtime hours
- Negative overtime hours
- Employee IDs that do not exist

Data Storage

This beginner version stores employee information in a Python list while the program is running.

The data is therefore temporary. When the program is closed, the current employee data is lost.

A future version can add SQLite or file-based storage for permanent records.

Project Structure

A simple submission can contain:

```
university_payroll/
■
■■■ university_payroll.py
■■■ README.md
```

Main Python Concepts Demonstrated

Variables

The program stores employee and salary information using variables.

Lists

The `employees` list stores employee records.

Dictionaries

Each employee is represented using a dictionary containing:

```
id
name
department
role
type
salary
overtime
```

Functions

The program is divided into functions such as:

```
add_employee()
view_employees()
find_employee()
add_overtime()
calculate_payroll()
generate_payslip()
search_employee()
remove_employee()
payroll_report()
main()
```

Conditional Statements

The program uses `if`, `elif`, and `else` to process menu selections and validate information.

Loops

A `while` loop keeps the main menu running until the user selects Exit.

Exception Handling

`try` and `except` are used to handle invalid numeric input.

Limitations

This is an educational beginner-level application. It currently:

• Runs only in the terminal.

• Stores data only during program execution.

• Does not have login/authentication.

• Does not use a permanent database.

• Does not generate PDF payslips automatically.

• Does not connect to a university ERP system.

Future Enhancements

Possible improvements include:

• SQLite database

• Employee login

• Administrator login

• Attendance management

• Leave management

• Monthly payroll history

• CSV export

• PDF payslip generation

• Configurable tax rates

• Graphical or web interface

• Role-based access control

Testing Checklist

Before submission, manually test:

• [ ] Add a Faculty employee

• [ ] Add a Staff employee

• [ ] View employees

• [ ] Search by name

• [ ] Search by department

• [ ] Add overtime

• [ ] Generate a payslip

• [ ] Generate payroll report

• [ ] Remove an employee

• [ ] Enter invalid salary

• [ ] Enter invalid employee ID

• [ ] Exit the application

Academic Project

Course: Python Essentials

Project: University Payroll Management System

Execution: Command line / terminal

Language: Python

Author

Student Name: _____

Registration Number: _____

University: _____

Course: Python Essentials